Content:

Fundamentals of Thermodynamics: *Introduction to thermodynamics: Quasi – static process. Zeroth law of thermodynamics, Liquid, gas, resistance thermometers. Joule's experiment (equivalence between heat and work), Numerical problems.*

First law of thermodynamics*, work done in thermodynamic quasi static processes, Isothermal process, adiabatic process and cyclic process, Application of first law of thermodynamics for both closed system and Steady State System. Numerical problems.*

Thermodynamics:

Thermodynamics is an axiomatic science which deals with the relations among heat, work and properties of system which are in equilibrium. It describes state and changes in state of physical systems.

System:

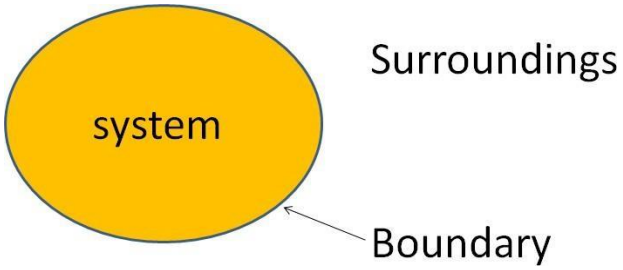
A thermodynamic system is defined as a quantity of matter or a region in space which isselected for the study.

Surroundings:

The mass or region outside the system is called surroundings.

Boundary:

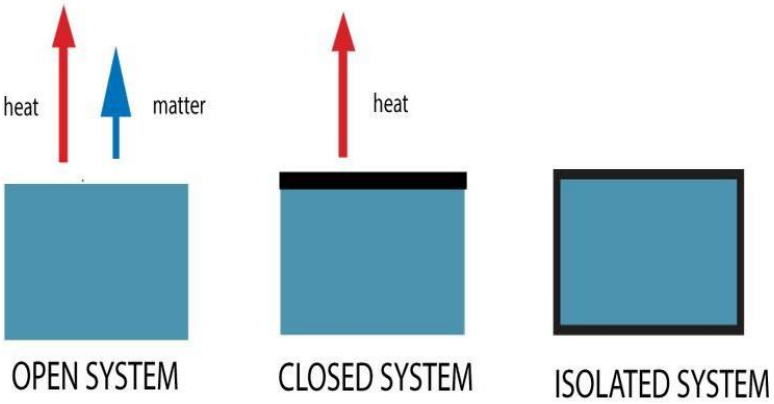
The real or imaginary surfaces which separates the system and surroundings is called boundary. The real or imaginary surfaces which separates the system and surroundings iscalled boundary.



Types of thermodynamic system

On the basis of mass and energy transfer the thermodynamic system is divided into threetypes.

1. Closed system
2. Open system
3. Isolated system



Closed system: A system in which the transfer of energy but not mass can take place acrossthe boundary is called closed system. The mass inside the closed system remains constant.

For example: Boiling of water in a closed vessel. Since the water is boiled in closed vessel so the mass of water cannot escape out of the boundary of the system but heat energy continuously entering and leaving the boundary of the vessel. It is an example of closed system.

Open system: A system in which the transfer of both mass and energy takes place is called an open system. This system is also known as control volume.

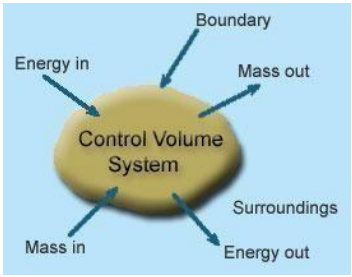
For example: Boiling of water in an open vessel is an example of open system because the water and heat energy both enters and leaves the boundary of the vessel.

Isolated system: A system in which the transfer of mass and energy cannot takes place is called an isolated system. For example: Tea present in a thermos flask. In this the heat and the mass of the tea cannot cross the boundary of the thermos flask. Hence the thermos flak is an isolated system.

Control Volume:

- It's a system of fixed volume.
- This type of system is usually referred to as "open system" or a "control volume"
- Mass transfer can take place across a control volume.
- Energy transfer may also occur into or out of the system.
- Control Surface- It's the boundary of a control volume across which the transfer of both mass and energy occur.
- The mass of a control volume (open system) may or may not be fixed.

- When the net influx of mass across the control surface equals zero then the mass ofthe system is fixed and vice-versa.
- The identity of mass in a control volume always changes unlike the case for a controlmass system (closed system).
- Most of the engineering devices, in general, represent an open system or controlvolume.



Example:

Heat exchanger - Fluid enters and leaves the system continuously with the transfer of heatacross the system boundary.
Pump - A continuous flow of fluid takes place through the system with a transfer of mechanical energy from the surroundings to the system.

Microscopic View or Study:

- The approach considers that the system is made up of a very large number of discrete particles known as molecules. These molecules have different velocities are energies. The values of these energies are constantly changing with time. This approach to thermodynamics, which is concerned directly with the structure of the matter, is known as statistical thermodynamics.
- The behavior of the system is found by using statistical methods, as the number of molecules is very large. So advanced statistical and mathematical methods are needed to explain the

changes in the system.

- The properties like velocity, momentum, impulse, kinetic energy and instruments cannot easily measure force of impact etc. that describe the molecule.
- Large numbers of variables are needed to describe a system. So the approach is complicated.

Macroscopic View or Study:

- In this approach a certain quantity of matter is considered without taking into account the events occurring at molecular level. In other words, this approach to thermodynamics is concerned with gross or overall behavior. This is known as classical thermodynamics.
- The analysis of macroscopic system requires simple mathematical formula.
- The value of the properties of the system are their average values. For examples consider a sample of gas in a closed container. The pressure of the gas is the average value of the pressure exerted by millions of individual molecules.
- In order to describe a system only a few properties are needed.

Sl. No	Macroscopic Approach	Microscopic Approach
1	In this approach a certain quantity of matter is considered without taking into account the events occurring at molecular level.	The matter is considered The effect of molecular motion is considered.
2	Analysis is concerned with overallbehavior of the system.	The Knowledge of the structure ofmatter is essential in analyzing the behavior of the system.
3	This approach is used in the study of classical thermodynamics.	This approach is used in the study of statistical thermodynamics.
4	A few properties are required to describe the system.	Large numbers of variables are required to describe the system.
5	The properties like pressure, temperature, etc. needed to describe the system, can be easily measured.	The properties like velocity, momentum, kinetic energy, etc. needed to describe the system, cannot be measured easily.
6	The properties of a system is given by its average values.	The properties are defined for each molecule individually.

7	This approach requires simple mathematical formulas for analyzing the system.	No. of molecules are very large so it requires advanced statistical and mathematical method to explain any change in the system.

Thermodynamic Equilibrium:

A thermodynamic system is said to exist in a state of thermodynamic equilibrium when no change in any macroscopic property is registered if the system is isolated from its surroundings.
An isolated system always reaches in the course of time a state of thermodynamic equilibrium and can never depart from it spontaneously.
Therefore, there can be no spontaneous change in any macroscopic property if the system exists in an equilibrium state. A thermodynamic system will be in a state of thermodynamic equilibrium if the system is the state of Mechanical, Chemical and Thermal equilibrium.

- Mechanical equilibrium: The criteria for Mechanical equilibrium are the equality ofpressures.
- Chemical equilibrium: The criteria for Chemical equilibrium are the equality ofchemical potentials.
- Thermal equilibrium: The criterion for Thermal equilibrium is the equality oftemperatures.

Property:

Thermodynamic Properties may be extensive or intensive.

Intensive properties: The properties which are independent of the mass of the system. For example: Temperature, pressure and density are the intensive properties.

Extensive properties: The properties which depend on the size or extent of thesystem are called extensive properties.
For example: Total mass, total volume and total momentum.

Process:

When the system undergoes change from one thermodynamic state to final state due change in properties like temperature, pressure, volume, the system is said to have undergone thermodynamic process.

Various types of thermodynamic processes are: Isothermal process, Adiabatic process, Isochoric process, Isobaric process and Reversible process.

Cycle:

Thermodynamic cycle refers to any closed system that undergoes various changes due to temperature, pressure, and volume, however, its final and initial state are equal. This cycle is important as it allows for the continuous process of a moving piston seen in heat engines and the expansion/compression of the working fluid in refrigerators, for example. Without the cyclical process, a car wouldn't be able to continuously move when fuel is added, or a refrigerator would not be able to stay cold. Visually, any thermodynamic cycle will appear as a closed loop on a pressure volume diagram.
Examples are Otto cycle, Diesel Cycle, Brayton Cycle etc.

Reversibility:

Reversibility, in thermodynamics, a characteristic of certain processes (changes of a system from an initial state to a final state spontaneously or as a result of interactions with other systems) that

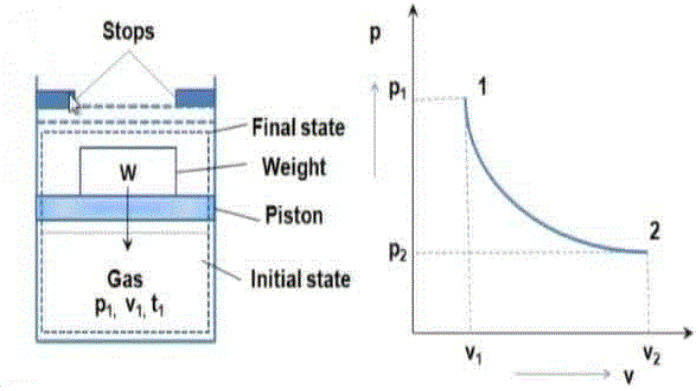
can be reversed, and the system restored to its initial state, without leavingnet effects in any of the systems involved.

An example of a reversible process would be a single swing of a frictionless pendulum from one of its extreme positions to the other. The swing of a real pendulum is irreversible because a small amount of the mechanical energy of the pendulum would be expended in performing work against frictional forces. Restoration of the pendulum to its exact starting position would require the supply of an equivalent amount of energy from a second system, such as a compressed spring in which an irreversible change of state would occur.

Quasi static process:

When a process is processing in such a way that system will be remained infinitesimally close with equilibrium state at each time, such process will be termed as quasi static process or quasi equilibrium process.

In simple words, we can say that if system is going under a thermodynamic process through succession of thermodynamic states and each state is equilibrium state then the process willbe termed as quasi static process.



We will see one example for understanding the quasi-static process, but let us consider one simple example for better understanding of quasi static process. If a person is coming down from the roof to the ground floor with the help of a ladder then it could be considered as quasi static process. But if he jumps from roof to ground floor then it will not be a quasi-static process.

Weight placed over the piston is just balancing the force which is exerted in upward direction by gas. If we remove the weight from the piston, system will have unbalanced force and piston will move in upward direction due to force acting over the piston in upward direction by the gas.

Irreversible Process:

The irreversible process is also called the natural process because all the processes occurring in nature are irreversible processes. The natural process occurs due to the finite gradient between the two states of the system. For instance, heat flow between two bodies occurs due to the temperature gradient between the two bodies; this is in fact the natural flow of heat. Similarly, water flows from high level to low level, current moves from high potential to low potential, etc.

- In the irreversible process the initial state of the system and surroundings cannot berestored from the final state.
- During the irreversible process the various states of the system on the path ofchange from initial state to final state are not in equilibrium with each other.
- During the irreversible process the entropy of the system increases decisively and itcannot be reduced back to its initial value.

The phenomenon of a system undergoing irreversible process is called asirreversibility.

Causes of Irreversibility:

Friction: Friction is invariably present in real systems. It causes irreversibility in the process as work done does not show an equivalent rise in the kinetic or potential energy of the system. The fraction of energy wasted due to frictional effects leads to deviation from reversible states.

Free expansion: Free expansion refers to the expansion of unresisted type such as expansion in a vacuum. During this unresisted expansion the work interaction is zero, and without the expense of any work, it is not possible to restore initial states. Thus, free expansion is irreversible.

Heat transfer through a finite temperature difference: Heat transfer occurs only when there exists temperature difference between bodies undergoing heat transfer. During heat transfer, if heat addition is carried out in a **finite number of steps** then after every step the **new state shall be a non-equilibrium state.**

Non equilibrium during the process: Irreversibilities are introduced due to lack of thermodynamic equilibrium during the process. Non-equilibrium may be due to mechanical inequilibrium, chemical inequilibrium, thermal inequilibrium, electrical inequilibrium, etc. and irreversibility is called mechanical irreversibility, chemical irreversibility, thermal irreversibility, electrical irreversibility respectively. Factors discussed above are also causing non-equilibrium during the process and therefore make process irreversible.

Heat:

It is the energy in transition between the system and the surroundings by virtue of the difference in temperature Heat is

energy transferred from one system to another solely by reason of a temperature difference between the systems. Heat exists only as it crosses the boundary of a system and the direction of heat transfer is from higher temperature to lower temperature. For thermodynamics sign convention, heat transferred to a system is positive; Heat transferred from a system is negative.

Work:

Thermodynamic definition of work: Positive work is done by a system when the sole effectexternal to the system could be reduced to the rise of a weight.
Work done by the system is positive and work done on the system is negative.

Types of work interaction:

- Expansion and compression work (displacement work)
- Work of a reversible chemical cell
- Work in stretching of a liquid surface
- Work done on elastic solids
- Work of polarization and magnetization

***Point and Path functions**:*

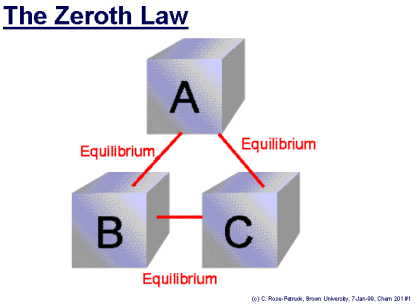
- Point function does not depend on the history (or path) of the system. It onlydepends on the state of the system.
- Examples of point functions are: temperature, pressure, density, mass, volume, enthalpy, entropy, internal energy etc.
- Path function depends on history of the system (or path by which system arrived at agiven state).

- Examples for path functions are work and heat.
- Path functions are not properties of the system, while point functions are propertiesof the system.
- Change in point function can be obtained by from the initial and final values of the function, whereas path has to defined in order to evaluate path functions.

Zeroth Law of Thermodynamics:

The Thermodynamics Zeroth Law states that if two systems are at the same time in thermal equilibrium with a third system, they are in equilibrium with each other.

If an object with a higher temperature comes in contact with an object of lower temperature, it will transfer heat to the lower temperature object. The objects will approach the same temperature and in the absence of loss to other objects, they will maintain a single constant temperature. Therefore, thermal equilibrium is attained.



Principles of Thermometry:

Thermometry is the science and practice of temperature measurement. Any measurable change in a thermometric probe (example: the dilation of a liquid in a capillary tube, variation of electrical resistance of a conductor, of refractive index of a transparent material, and so on) can be used to mark temperature levels, that should later be calibrated against an internationally agreed unit if the measure is to be related to other thermodynamic variables.

Liquid (in glass) thermometers:

A liquid-in-glass thermometer measures temperature based on the thermal expansion of mercury or spirit/ alchohol in a glass container. The boiling point of mercury is 356.72 °C, and its melting point is -38.86 °C. The boiling point of methyl alcohol is 64.65 °C and its melting point is -97.78 °C. Because mercury has low thermal capacity, high heat conductivity, inertness in relation to a glass capillary tube and a high boiling point, it is an ideal thermometric liquid except for its relatively high melting point. Accordingly, mercury thermometers are used for ordinary meteorological observations, and spirit thermometers are used for measurements involving temperatures below the melting point of mercury.

Thermometry is sometimes split in metrological studies in two subfields: contact thermometry and noncontact thermometry. As there can never be complete thermal uniformity at large, thermometry is always associated to a heat transfer problem with some space-time coordinates of measurement, given rise to time-series plots and temperature

The liquid-in-glass thermometer consists of a capillary glass tube with a bulb at one end filled with a thermometric liquid,

vacuumed and sealed. By reading the position of the liquid level on a scale, a temperature value can be obtained. Designs can be classified as either the sheathed type or the unsheathed type. A sheathed thermometer consists of a bulb, a slender capillary glass tube connected to it, a milky-white scale plate attached to the capillary tube, and an outer glass tube that encloses them. An unsheathed thermometer consists of a thick-walled capillary glass tube with a scale marked directly on it.

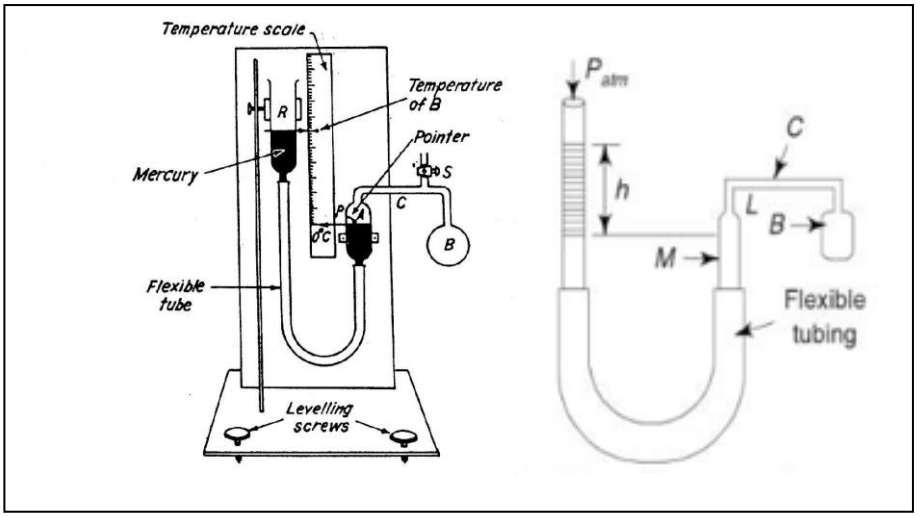
The advantages of liquid-in-glass thermometers are their simple design, simple observation method and the possibility of temperature measurement anywhere as they require no electric power for operation. The disadvantage is that careful handling is required because the glass material is fragile. Several types of liquid-in-glass thermometers are used to measure maximum temperature, minimum temperature and soil temperature in addition to ordinary air temperature.

Constant Volume Gas Thermometers

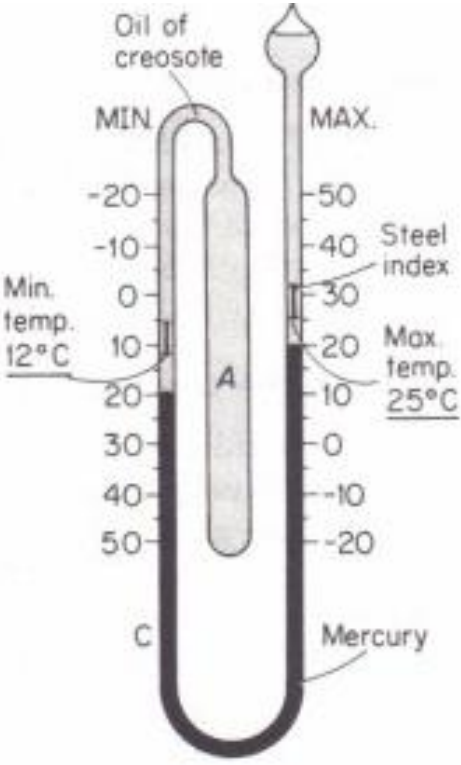
This thermometer consists of a capillary tube (C), which connects thermometer bulb with a U-tube manometer. A small amount of helium-gas is contained in the bulb 'B'. The left limb of the manometer is kept open to atmosphere and can be moved vertically. The mercury level on the right limb can be adjusted so that it just touches lip 'L' of the capillary. The pressure of the gas in the bulb is used as a thermometric property and is given by $P = P_{atm} + \rho_m gh$. In this formula, P_{atm} is the atmospheric pressure and ρ_m is the density of Mercury. When the bulb is brought in contact with the system whose temperature I to be measured, it comes in thermal equilibrium with the system. The gas in the bulb will be heated and expanded and it pushes the mercury column downwards in the right limb. This raises the mercury column in the left column. The flexible tube is adjusted so that the mercury again touches the lip 'L'. The

difference in mercury level 'h' is recorded and the Pressure 'P' of the gas in the bulb is calculated. Since the volume of gas in the bulb is kept constant, from ideal gas equation PV = mRT, since m ,R is constant we get $V\Delta P = mR\Delta T$. Thus we get the important relation that ΔT is proportional to ΔP. Hence we get :

$\frac{T}{T_{tp}} = \frac{P}{P_{tp}}$. Thus $T = \frac{P}{P_{tp}} T_{tp}$ that is $T = 273.16 \frac{P}{P_{tp}}$ where T_{tp} = Triple point temperature of water and P_{tp} is the triple point pressure of water.



Six's Maximum and Minimum thermometer (invented by James Six in 1782)



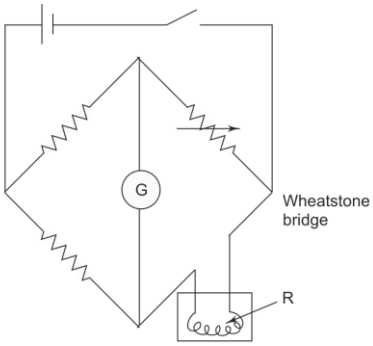
It measures the maximum and minimum temperatures reached since the thermometer was last read. Minimum temperature occurs during the night while the maximum temperature during the day. It consists of a fairly large cylindrical bulb A which originally contained alcohol, although oil of creosote is now more generally used. This is connected by a U-shaped stem to a second bulb nearly full of the same liquid. The bend of the U contains a thread of mercury. Two scales are provided, one against each limb of the tube so that the temperature may be read against either of the mercury levels. Resting on each of the mercury surfaces are small steel indexes provided with light springs to hold them in position in the stem. Expansion or contraction of the liquid in A causes a movement of the mercury thread. Consequently, one or other index is pushed forward by the mercury and left in the extreme position reached. Thus, the lower end of the index on the left indicates the minimum and that on the right the maximum temperature attained. After

readings have been taken a small magnet is used to bring the indexes back into contact with the mercury.

Resistance thermometers

Electrical resistance thermometer, semiconductor thermometer (thermistor) and thermocouple thermometer are three types of electrical thermometers that are used commonly. Here we focus on Platinum-resistance thermometer as an example of electrical resistance thermometer.

Platinum resistance thermometer employs platinum characteristics which changes resistance according to the temperature. It allows us to obtain temperature by measuring electrical resistance. High purity platinum is used since contaminants greatly affect resistance. Sensor for meteorological observation is made with thin sheet of mica or porcelain wrapped with platinum wire, and it is placed in stainless protective tube which has excellent thermal conductivity, is corrosion resistant and is made complete water proof. Diagram of sensor and connection of platinum resistant thermometer is shown in Figure below.

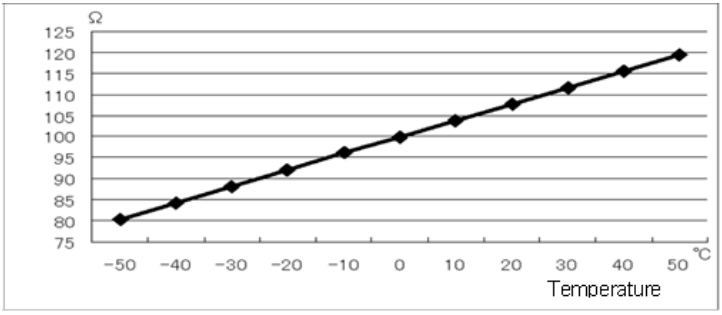


In the resistance thermometer, the change in resistance of a metal wire due to its change in temperature is the thermometric property. The wire, frequently platinum, may be incorporated in a Wheatstone bridge circuit. The platinum resistance thermometer

measures temperature to a high degree of accuracy and sensitivity, which makes it suitable as a standard for the calibration of other thermometers.

In a restricted range, the following quadratic equation is used $\mathbf{R} = \boldsymbol{R_0}(1 + At + Bt^2)$, where where R_0 is the resistance of the platinum wire when it is surrounded by melting ice and A and B are constants.

Resistance change of platinum due to temperature is converted to electrical signals (current or voltage signal) by converter. Then the signal is sent to indicator or recorder and displayed or processed as atmospheric temperature. An example of the relation between temperature and resistance of platinum resistance thermometer is shown in Figure below.

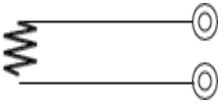


The advantage of platinum resistance thermometer is the electric measurement which allows remote measurement and automated observation. The disadvantage is necessity of ensuring stable power supply to platinum resistance.

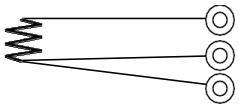
There are 2-conductor system, 3-conductor system and 4-conductor system in the method of inside conductor connection. 2-conductor system is impractical because it is unable to remove the effect of conductor resistance. 3-conductor system consists of 2

conductors on one terminal of a resistance element and 1 conductor on the other terminal. This method enables to remove effect of conductor resistance. However, this is based on the premise that conductor has identical material, length and electrical resistance and that the temperature distribution must be the same. 4-conductor system consists of 2 conductors connected on each terminal of resistant element. This method can remove effect of conductor resistance. 4-conductor system has the highest accuracy as a thermometer.

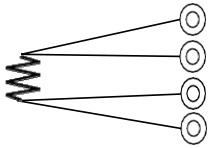
Diagram of inside conductor connection



2 conductor system



3 conductor system



4 conductor system

Scales of Temperature:

There are three temperature scales in use Fahrenheit, Celsius and Kelvin. Fahrenheit temperature scale is a scale based on 32 for the freezing point of water and 212 for the boiling point of water, the interval between the two being divided into 180 parts.

The conversion formula for a temperature that is expressed on the Celsius (C) scale to its Fahrenheit (F) representation is:

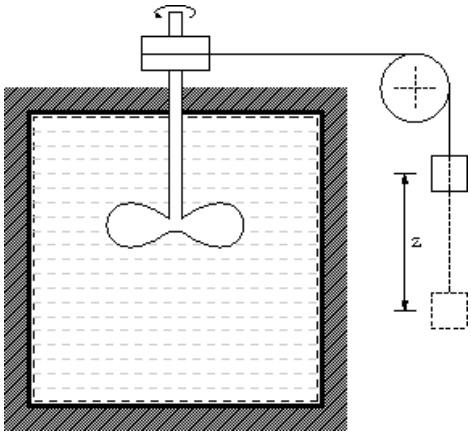
$F = 9/5\ C + 32.$

Celsius temperature scale also called centigrade temperature scale, is the scale based on 0 for the freezing point of water and 100 for the boiling point of water.

Kelvin temperature scale is the base unit of thermodynamic temperature measurement in the International System (SI) of measurement. It is defined as 1/ 273.16 of the triple point (equilibrium among the solid, liquid, and gaseous phases) of pure water.

Joule's experiment:

James P. Joule carried out his famous experiment; he placed known amounts of water, oil, and mercury in an insulated container and agitated the fluid with a rotating stirrer. The amounts of work done on the fluid by the stirrer were accurately measured, and the temperature changes of the fluid were carefully noted. He found for each fluid that a fixed amount of work was required per unit mass for every degree temperature rise caused by the stirring, and that the original temperature of the fluid could be restored by the transfer of heat through simple contact with a cooler object. In this experiment you can conclude there is a relationship between heat and work or in other word heat is a form of energy. This connection between the work done on the liquid and the rise in temperature of the liquid can be written as : $Mgz = \Delta Q = mc\Delta\theta$, where M is the mass of the body sliding down by distance z and m is the mass of the liquid with specific heat capacity c.



Internal Energy

In the Joule experiment what happens to energy between the time it is added to water as work and the time it is extracted as heat? Logic

suggests that this energy is contained in water in another form called the Internal energy.

Internal energy refers to energy of the molecules of a substance which undergoes ceaseless motion and thereby has kinetic energy. The addition of heat to a substance increases this molecular activity, and thus causes an increase in its internal energy. Work done on the substance can have the same effect, as was shown by Joule. Internal energy cannot be directly measured since there are no internal energy meters. As a result, its absolute value is unknown. However,this is not a disadvantage in thermodynamic analysis, because only changes in internal energy are required.

First Law of Thermodynamics:

During a thermodynamic cycle, since the system undergoes a cyclic process, the cyclic integral of heat added is equal to integral of work done. The First law of thermodynamics can be written in the form,

$$0 = \oint (dQ - dW)$$

Equation dU = dQ – dW is a corollary to the first law of thermodynamics. It shows that there exists a property called the internal energy (U) of the system, such that a change in its value is equal to the difference in heat entering and work leaving the system. The First law of thermodynamics states that energy is neither created nor destroyed. Thus, the total energy of the universe is a constant. However, energy can certainly be transferred from one form to another form. The 1st law of thermodynamics can be mathematically stated as follows:

$$\oint Q = \oint W$$

Corollary 1:

There exists property of closed system; the change in value of this property during the process is given by the difference between heat supplied and work done.

$$dU = dQ - dW$$

Here U is property of system and is called as total energy that includes internal energy, kinetic energy, potential energy, electrical energy, magnetic energy, chemical energy, etc.

Corollary 2:

For the isolated system, heat and work both interactions are absent ($dQ = 0$, $dW = 0$) and U is a constant. Energy can neither be created nor be destroyed; but it can be converted from one form to other.

Corollary 3:

A perpetual motion machine of first kind (PMM1) is impossible as it violates the 1st law of thermodynamics. PMM1 is a hypothetical machine which can produce useful work without any source or which can produce more energy than consumed. First law of thermodynamics states that energy can be neither created or nor destroyed. A perpetual motion machine 1 violates the law by creating energy. All processes in universe comply with first and second law of thermodynamics. Therefore, PPM1 does not and can never exist.

e.g.: An electric heater that consumes 1KW of electricity cannot produce more than 1KW heat.

Work dine in Ideal gas process

In this section we investigate the ideal-gas process from the point of view of work done.

We have seen that the work done by a force F_s on a system as it moves from position s_i to position s_f is defined as the integral:

$$W = \int_{si}^{sf} Fsds$$

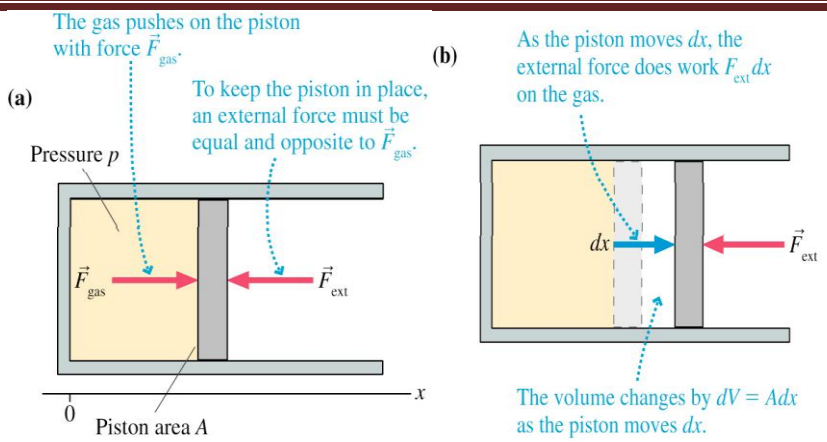
Let us calculate the work done on a gas as it expands or compresses. Suppose the gas is in a container closed at one end. At the other end a tightly- fitting moveable piston enables us to apply an external force $F_{ext} = -F_{gas} = -pA$.

Suppose the piston moves a small distance dx to the right shown in Figure below. As it does so, the external force (the environment) does an amount of work on the piston and therefore on the gas of :

$$dW = F_{ext}\, dx = -pAdx$$

If dx is positive (the gas expands) then dW is negative; if dx is negative (the gas is compressed) then dW is positive. As the piston moves the distance dx, the volume of the gas changes by an amount $dV = Adx$.

Illustration of the work done by an external force on a piston, and therefore on a gas is as below



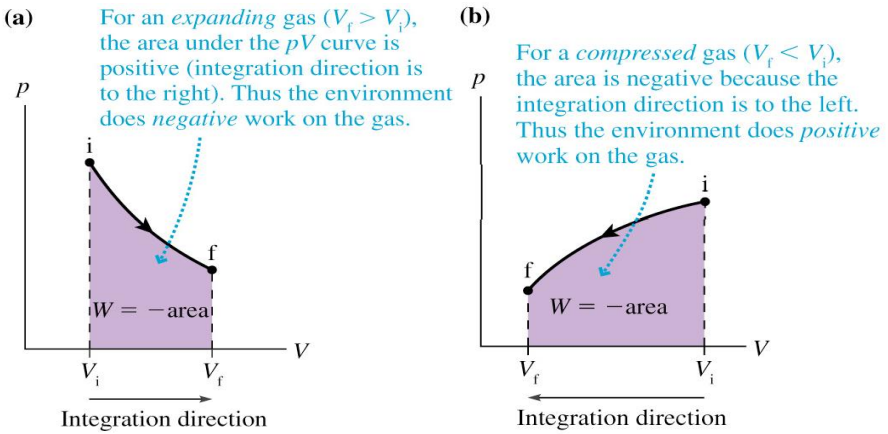
Consequently, we can write the above equation for the work done as: dW = -pdV.

If in the process the volume changes from V_i to V_f the total work done by F_{ext} is the integral

$$W = -\int_{Vi} pdv$$

To evaluate this integral we need to know the dependence of p on V (the function $p(V)$). What is the geometrical meaning of work done by the external force ? Eq above states that the work done on a gas is the negative of the area under the gas's pV curve as the volume changes from V_i to V_f. Fig (a) below shows gas expansion. The area under the pV curve is positive. This means that the environment does a negative amount of work. Figure (b) below shows a contraction. The area is negative (the integration is performed from a larger to a smaller volume) so the work done by

the environment on the gas is positive. Let us consider a numerical example of these geometrical interpretations.



The work done on a gas by an external force is the negative of the area under the gas's pV curve. The direction of the process is shown by the arrow on the trajectory.

Isochoric process

In the isochoric process (Fig a) the volume does not change. The area under the curve is zero and consequently, the work done is zero: $W = 0$ (isochoric process)d

Isobaric process

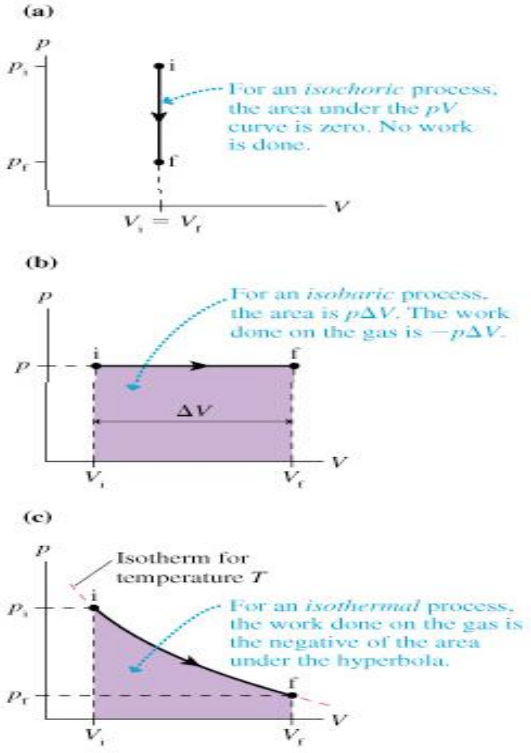
In the isobaric process (Fig b) the volume changes from V_i to V_f. The area under the curve andthe work done is $W = -p\Delta V$. If ΔV is positive (the gas expands) then W is negative. If ΔV is negative (the gas is compressed) then W is positive.

Isothermal process

An isothermal process is shown in (Fig c). Substituting $p = nRT/V$ in the integral for work done, as the volume changes from V_i to V_f.

$$W = -\int_{V_i}^{V_f} pdV = -(nRT)\int_{V_i}^{V_f}(dV/V) = -nRT\,[\ln(Vf) - \ln(Vi)]$$

$$= -nRT\ln(V_f/V_i) = -P_iV_i\ln(V_f/V_i) = -P_fV_f\ln(V_f/V_i)$$



Adiabatic Expansion of Gas:

The process in which no heat enters the system undergoing the change from the surroundings neither does any heat leave the system to the surroundings is an Adiabatic process. The corresponding changes in volume and pressure are aid to be Adiabatic changes.

Thus in terms of the 1st Law of Thermodynamics, we can say that for a thermodynamic process, dQ = dU + dW = 0.

Derivation of Adiabatic equation of state : PV^{Υ} = Constant.

$dU = mC_v dT$, $dW = PdV$. Thus $dQ = mC_v dT + PdV = 0$.

So we arrive at : $mC_v dT = - PdV$.

But $PV = mRT$ is the gas equation of state. Hence we get :
$PdV + VdP = mRdT$. So, $dT = (P/mR)dV + (V/mR)\, dP$.

Thus $dQ = 0 = mC_v [\, (P/mR)dV + (V/mR)\, dP\,] + PdV$

So, $C_v [\, (P/R)dV + (V/R)\, dP\,] + PdV = 0$

$C_v [\, PdV + V\, dP\,] + RPdV = 0$

$(C_v + R)PdV + C_v\, VdP = 0$

But $C_p - C_v = R$, which means $C_p = C_v + R$

Thus we get : $C_p PdV + C_v\, VdP = 0$

Dividing throughout by $C_p PV$, we get :

$dV/V + (C_v/C_p)(dP/P) = 0$

$(C_p/ C_v)\, dV/V + dP/P = 0$

$\Upsilon\, (dV/V) + (dP/P) = 0$

Hence, after integration, we get PV^{Υ} = Constant

Work done in Adiabatic Expansion of Gases :

$PV^{\Upsilon} = K$ (K is the constant). This means that $P = K/V^{\Upsilon}$

$dW = PdV = (K/V^{\Upsilon})dV$.

After integrating the above equation, we get : $W = \int dW = K\int V^{-\Upsilon}\, dV$

$$\mathbf{W = (K/(1 - \Upsilon))V^{(1-\Upsilon)}}$$

For adiabatic expansion, the initial and final volumes are V_i and V_f, where $V_f > V_i$

So work done in adiabatic expansion is $W = (K/(1 - \Upsilon))[V_f^{(1-\Upsilon)} - V_i^{(1-\Upsilon)}] = (K/(\Upsilon - 1))[V_i^{(1-\Upsilon)} - V_f^{(1-\Upsilon)}]$

$W = (1/(\Upsilon - 1))[(K/V_i^{(\Upsilon-1)}) - (K/V_f^{(\Upsilon-1)})]$

Putting $K = P_iV_i^{\Upsilon}$ in the first term and $K = P_fV_f^{\Upsilon}$ in the second term, we get :

$W = (1/(\Upsilon - 1))[(P_iV_i^{\Upsilon}/V_i^{(\Upsilon-1)}) - (P_fV_f^{\Upsilon}/V_f^{(\Upsilon-1)})] = [P_iV_i - P_fV_f]/(\Upsilon - 1)$

Thus $W = [P_iV_i - P_fV_f]/(\Upsilon - 1)$

Limitations of the First Law:

- First law does not provide a clear idea about the direction of absorption or evolution ofheat.
- The first law of thermodynamics merely indicates that in any process there is a transformation between the various forms of energy involved in the process but has nothing to say regarding the feasibility of such transformation.
- First law does not provide any information regarding the direction in which processes will take place or whether it is a spontaneous or a non-spontaneous process.

Thermal Reservoir:

A thermal reservoir is a large system (very high mass x specific heat value) from which a quantity of energy can be absorbed or added as heat without changing its temperature. The atmosphere and sea are examples of thermal reservoirs. Any physical body whose thermal energy capacity is large relative to the amount of

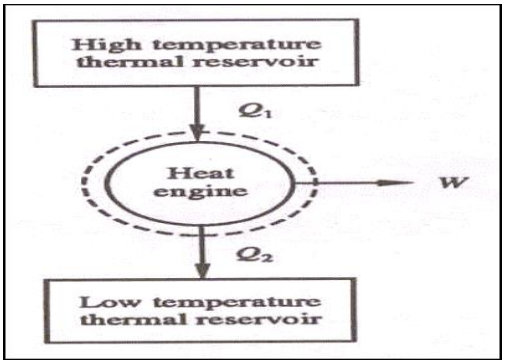
energy it supplies or absorbs can be modelled as a thermal reservoir. A reservoir that supplies energy in the form of heat is calleda source and one that absorbs energy in the form of heat is called a sink.

Heat Engine:

It is a cyclically operating device which absorbs energy as heat from a high temperature reservoir, converts part of the energy into work and rejects the rest of the energy as heat to a thermal reservoir at low temperature.

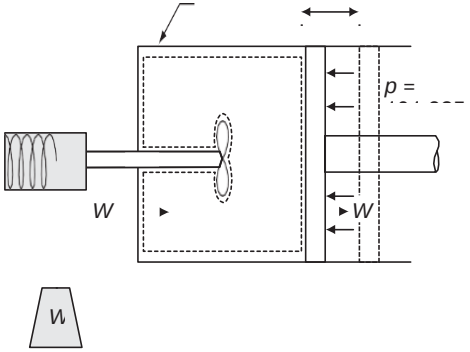
The working fluid is a substance, which absorbs energy as heat from a source, and rejects energy as heat to a sink.

Schematic representation of Heat Engine

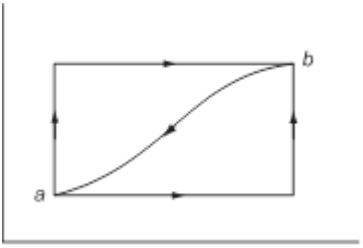


SL No	Sample short answer questions	CO
1	What is meant by thermodynamic equilibrium? Explain with the help of examples.	1
2	Write about constant volume gas Thermometer? Why it is preferred over a constant pressure gas Thermometer.	1
3	Distinguish between macroscopic and microscopic point of view	1
4	Discuss Quasi Static process, what are its characteristics	2
5	Discuss First law of thermodynamics, explain Joule's experiment	2
6	The pressure of as in a pipe line is measured with a mercury manometer having one limb open to the atmosphere. If the difference in the height of mercury in the two limbs is 562 mm, calculate the gas pressure. The barometer reads 761 mm Hg, the acceleration due to gravity is p_0 9.79 m/s^2, and the density of mercury is 13,640 kg/m^3. **Solution:** At the lower liquid plane in the manometer, we have $p = p_o + \rho gz$ Now $p_o = \rho g z_o$ where z_0 is the barometric height, ρ the density of mercury and p_0 theatmospheric pressure. Therefore $p = \rho g\ (z + z_0)$ = 13,640 kg/m^3 × 9.79 m/s^2 (0.562 + 0.761) m = 177 × 10^3 N/m^2 = 177 kPa = 1.77 bar = 1.746 atm	3
7	A turbine is supplied with steam at a gauge pressure of 1.4 MPa. After expansion in the turbine the steam flows into a condenser which is maintained at a vacuum of 710 mm Hg.	2

	The barometric pressure is 772 mm Hg. Express the inlet and exhaust steam pressure in pascals (absolute). Take the density of mercury as 13.6×10^3 kg/m^3 **Solution:** The atmospheric pressure p $= \rho gz = 13.6 \times 10^3$ kg/m^3 $\times$ 9.81 m/s^2 $\times$ 0.772 m $= 1.03 \times 10^5$ Pa Inlet steam pressure = $(1.4 \times 10^6) + (1.03 \times 10^5)$ Pa $= 15.03 \times 10^5$ Pa = 1.503 MPa Condenser pressure = (0.772 − 0.710) m $\times$ 9.81 m/s^2 $\times$ 13.6×10^3 kg/m^3 $= 0.827 \times 10^4$ Pa = 8.27 kPa	
8	When the valve of the evacuated bottle is opened, atmospheric air rush pressure is 101.325 kPa, and 0.3 m^3 of air (measured at atmospheric bottle, calculate the work done by air. **Solution:** The displacement work done by air $W_d = \int_{Bottle} pdv + \int_{Free-air\ boundary} pdv$ $= 0 + p\Delta v$ $= 101.325 \times 0.6$ $= 60.8$ kJ Since the free-air boundary is contracting, the work done by the system is negative (ΔV being negative), andthe surroundings do positive work upon the system.	3

9	*A piston and cylinder machine containing a fluid system has a stirring device in the cylinder (Fig. Ex. 3.3). The piston is frictionless, and it is held down against the fluid due to the atmospheric pressure of 101.325 kPa. The stirring device is turned 10,000 revolutions with an average tor ue against the fluid of 1.275 mN. Meanwhile the piston of 0.6 m diameter*  Fig. Ex. **Solution:** Work done by the stirring device upon the system $W_1 = 2\pi TN$ $= 2\pi \times 1.275 \times 10{,}000$ Nm = 80 kJ This is negative work for the system. Work done by the system upon the surroundings $W_2 = (pA) \cdot L$ $= 101.325\ \text{kN} \times \pi\ (0.6)^2\ \text{m}^2 \times 0.80$ m = 22.9 kJ This is positive work for the system. Hence, the net work transfer for the system $W = W_1 + W_2 = -80 + 22.9 = -57.1$ kJ	3
1	If it is desired to melt aluminium with solid state specific heat	3

0	0.9 kJ kg K, latent heat 390 kJ kg, atomic weight 27, density in molten state 2400 kg m^3 and final temperature 700 C, find out how much metal can be melted per hour with the above kW rating. ther data are as in the above example. Also, find the mass of aluminium that the above furnace will hold. The melting point of aluminium is 660 C. **Solution:** Heat required per kg of aluminium = 0.9 (660 – 15) + 390 + (700 – 660)(29.93/27) = 580.5 + 390 + 44.3 = 1014.8 kJ Heat to be supplied = 1014.8/0.7 = 1449.7 kJ/kg With the given power, the rate at which aluminium can be melted = (2.17 x 1000 x 3600)/1449.7 kg/h = 5.39 tonnes/h Mass of aluminium that can be held in the above furnace = 2.18 × 2400 kg = 5.23 tonnes	
11	A stationary mass of gas is compressed without friction from an initial state of 0.3 m^3 and 0.105 MPa to a final state of 0.15 m^3 and 0.105 MPa, the pressure remaining constant during the process. There is a transfer of 37.6 kJ of heat from the gas during the process. How much does the internal energy of the gas change? **Solution:** First law for a stationary system in a process gives $Q = \Delta U + W$ Or $Q_{1-2} = U_2 - U_1 + W_{1-2}$ (1) Here $W_{12} = \int pdV = p(V_2 - V_1) = 0.105\ (0.15 - 0.30)$ MJ $= -15.75$ kJ $Q_{1-2} = -37.6$ kJ $\therefore$ Substituting in Eq. (1) -37.6 kJ $= U_2 - U_1 - 15.75$ kJ	3

	$\therefore U_2 - U_1 = -21.85$ kJ The internal energy of the gas decreases by 21.85 kJ in the process.	
12	When a system is taken from state a to state b, in Fig. Ex. 4.2, along path acb, 84 kJ of heat flow into the system, and the system does 32 kJ of work. a) How much will the heat that flows into the system along path adb be, if the work done is 10.5 kJ. b) When the system is returned from b to a along the curved path, the work done on the system is 21 kJ. Does the system absorb or liberate heat, and how much of theheat is absorbed or liberated, c) If $U_a = 0$ and $U_d = 42$ kJ, find the heat absorbed in the processes ad and db.  **Solution:** $Q_{acb} = 84$ kJ, $W_{acb} = 32$ kJ $Q_{acb} = U_b - U_a + W_{acb}$ $U_b - U_a = 84 - 32 = 52$ kJ, $Q_{adb} = U_b - U_a + W_{adb}$ $= 52 + 10.5 = 62.5$ kJ $Q_{b-a} = U_a - U_b + W_{b-a} = -52 - 21 = -73$ kJ. The system liberates 73 kJ of heat. $W_{adb} = W_{ad} + W_{db} = W_{ad} = 10.5$ kJ,	3

	$Q_{ad} = U_d - U_a + W_{ad} = 42 - 0 + 10.5 = 52.5$ kJ, $Q_{adb} = 62.5$ kJ = $Q_{ad} + Q_{db}$ $Q_{db} = 62.5 - 52.5 = 10$ kJ	
13	The internal energy of a certain substance is given by the following equation $u = 3.56\ pv + 84$ where u is given in kJ kg, p is in kPa, and v is in m^3 kg. A system composed of 3 kg of this substance expands from an initial pressure of 500 kPa and volume of 0.22 m^3 to a final pressure 100 kPa in a process in which pressure and volume are relatedby $pv^{1.2}$ = constant. *a* If the expansion is quasi static, find Q , ΔU, and W for the process. *b* In another process the same system expands according to the same pressure volume relationship as in part a , and from the same initial state to the same final state as in part a , but the heat transfer in this case is 30 kJ. Find the work transfer for this process. *c* Explain the difference in work transfer in parts a and b . **Solution:** $u = 3.56\ pv + 84$, $\Delta u = u_2 - u_1 = 3.56\ (p_2v_2 - p_1v_1)$, $\Delta U = 356\ (1 \times 0.845 - 5 \times 0.22)$ kJ $= -356 \times 0.255 = -91$ kJ For a quasi-static process, work done = ∫ pdV = (p2V2 – p1V1)/(1-ϒ) , where ϒ = 1.2 = (1x0.845 – 5x0.22)100/(1-1.2) = 127.5 kJ. So we get Q = ΔU + W = – 91 + 127.5 = 36.5 kJ b) Here Q = 30 kJ. Since the end states are the same, ΔU would remain the same as in (a). So, W = Q – ΔU = 30 – (– 91) = 121 kJ c) The work in (b) is not equal to∫ pdV since the process is	3

	not quasi-static.	
14	A piston cylinder device operates 1kg of fluid at 20atm pressure with initial volumeis 0.04m3. Fluid is allowed to expand reversibly following pV1.45=C. So that the volume becomes double. The fluid is cooled at constant pressure until the piston comes back. Determine the work done in each process?	3
15	When a stationary mass of gas was compressed without friction at constant pressure, its initial state of 0.4m3 and 0.105MPa was found to change to final state of 0.20m3 and 0.105MPa. There was a transfer of 42.5kJ of heat from the gas during theprocess. Determine the change in internal energy of the gas?	
16	0.44 kg of air at 1800C, expands adiabatically to 3times its original volume and during the process there is a fall in temperature to 150C. The work done during the processis 52.5kJ. Calculate Cp and Cv .	3
17	A new scale N of temperature is divided in such a way that the freezing point of ice is 1000N and the boiling point is 4000N. What is the temperature reading on this new scale when the temperature is 1500C? At what temperature both the Celsius and the new scale reading would be the same?	3
18	A piston-cylinder device operates 1 kg of fluid at 20 atm. Pressure. The initial volume is 0.04 m3. The fluid is allowed to expand reversibly following a process pV1.4 = constant so that the volume becomes double. The fluid is then cooled at a constant pressure until the piston comes back to the original position. Keeping the piston unaltered, heat is added reversibly to restore it to the initial pressure.	3

	Calculate the work done in the cycle.	
19	1.8 kg of oxygen at 48^0 C is mixed with 6.2 kg of nitrogen at the same temperature. Both oxygen and nitrogen are at the pressure of 102 k Pa before and after mixing. Find the increase in entropy	3
20	A mixture of ideal air and water vapour at a dbt of 220C and a total pressure of 730 mm Hg abs. has a temperature of adiabatic saturation of 150C. Calculate: i. the speci_c humidity in gms per kg of dry air. ii. the partial pressure of water vapour the relative humidity, and iv. enthalpy of the mixture per kg of dry air.	3
21	Atmospheric air at 1.0132 bar has a DBT of 30°C and WBT of 24°C. Compute, (i) The partial pressure of water vapour (ii) Specific humidity (iii) The dew point temperature (iv) Relative humidity (v) Degree of saturation (vi) Density of air in the mixture (vii) Density of the vapour in the mixture (viii) The enthalpy of the mixture.	3
22	A mixture of ideal gases consists of 3 kg of nitrogen and 6 kg of CO2 at a pressure of 250 kpa and a temperature of 230C. Find (a) the mole fraction of each constituent (b) equivalent weight of the mixture (c) equivalent gas constant of the mixture (d) partial pressures and partial volumes (e) volume and density of mixture Cp and Cv of mixture. If the mixture is heated at constant volume to 60C and the internal energy, enthalpy, and entropy of mixture. Find the changes in internal energy, enthalpy, and entropy of mixture if the heating is done at constant pressure.Take γ for CO2and N2 be 1.286 and 1.4.	3